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Visual and tactile confirmation of package presence for UAV aerial deliveries

Abstract

A technique for validating a presence of a package carried by an unmanned aerial vehicle (UAV) includes: capturing an image of a scene below the UAV with a camera mounted to the UAV and oriented to face down from the UAV; analyzing the image to identify whether the package is present in the image; and determining whether the package is attached to the UAV, via a tether extending from an underside of the UAV, based at least on the analyzing of the image.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8223200	12/2011	Benttine et al.	N/A	N/A
10730626	12/2019	Gil et al.	N/A	N/A
10885653	12/2020	Goja et al.	N/A	N/A
11407511	12/2021	Buchmueller et al.	N/A	N/A
2017/0267347	12/2016	Rinaldi et al.	N/A	N/A
2019/0197646	12/2018	Prager et al.	N/A	N/A
2020/0207474	12/2019	Foggia	N/A	G05D 1/0676
2020/0354083	12/2019	Priest	N/A	B64F 1/32

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111587444	12/2019	CN	N/A

OTHER PUBLICATIONS

Jordan, An Overview of Semantic Image Segmentation, https://www.jeremyjordan.me/semantic-segmentation, May 21, 2018. cited by applicant
Image segmentation, Wikipedia Article, https://en.wikipedia.org/w/index.php?title_image-Segmentation&oldid=1098934814, Jul. 18, 2022. cited by applicant
The International Search Report and Written Opinion mailed May 8, 2024, in corresponding PCT Application No. PCT/US2023/027539, 13 pages. cited by applicant

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Background/Summary

TECHNICAL FIELD
(1) This disclosure relates generally to unmanned aerial vehicles (UAVs), and in particular, relates to aerial package delivery systems using UAVs.
BACKGROUND INFORMATION
(2) An unmanned vehicle, which may also be referred to as an autonomous vehicle, is a vehicle

capable of travel without a physically present human operator. Various types of unmanned vehicles exist for various different environments. For instance, unmanned vehicles exist for operation in the air, on the ground, underwater, and in space. Unmanned vehicles also exist for hybrid operations in which multi-environment operation is possible. Unmanned vehicles may be provisioned to perform various different missions, including package delivery, exploration/reconnaissance, imaging, public safety, surveillance, or otherwise. The mission definition will often dictate a type of specialized equipment and/or configuration of the unmanned vehicle.
(3) Unmanned aerial vehicles (UAVs), commonly referred to as drones, hold the promise of providing an economical and environmentally friendly delivery service. As these UAVs start delivering packages to, or retrieving packages from, the general public, it will become increasingly important to ensure the correct package is attached to the UAV at the expected times throughout the various segments of an aerial delivery mission. As the public begins to accept aerial delivery services as reliable, the value of the packages entrusted to an aerial delivery service is expected to rise, which in turn places even more emphasis on ensuring packages are correctly attached. Ensuring that an erroneous package is not accepted or delivered is also a safety measure protecting recipient parties. Accordingly, techniques for effectively detecting and validating the presence of a package attached to a UAV providing an aerial delivery service are advantageous.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Non-limiting and non-exhaustive embodiments of the invention are described with reference to the following figures, wherein like reference numerals refer to like parts throughout the various views unless otherwise specified. Not all instances of an element are necessarily labeled so as not to clutter the drawings where appropriate. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating the principles being described.
(2) FIG. **1**A is a topside perspective view illustration of an unmanned aerial vehicle (UAV) adapted for aerial delivery missions, in accordance with an embodiment of the disclosure.
(3) FIG. **1**B is a bottom side plan view illustration of a UAV adapted for aerial delivery missions, in accordance with an embodiment of the disclosure.
(4) FIG. **2** is a functional block diagram illustrating subsystems of a UAV relevant for detecting and validating the presence of a package, in accordance with an embodiment of the disclosure.
(5) FIG. **3** is a flow chart illustrating a process for detecting and/or validating the presence of a package attached to a UAV using a combination of visual image analysis and tactile feedback from sensors, in accordance with an embodiment of the disclosure.
(6) FIG. **4**A illustrates an image of a scene below a UAV that includes the presence of a corner of a package in an expected location during a cruise segment of an aerial delivery mission, in accordance with an embodiment of the disclosure.
(7) FIG. **4**B illustrates how image analysis can be used to determine the presence and location of a package in an image, in accordance with an embodiment of the disclosure.
(8) FIG. **4**C illustrates how location and threshold number of contiguous pixels in an image can vary based upon a length of a tether and a current mission segment, in accordance with an embodiment of the disclosure.
DETAILED DESCRIPTION
(9) Embodiments of a system, apparatus, and method for detecting and validating packages attached to an unmanned aerial vehicle (UAV) are described herein. In the following description numerous specific details are set forth to provide a thorough understanding of the embodiments. One skilled in the relevant art will recognize, however, that the techniques described herein can be practiced without one or more of the specific details, or with other methods, components, materials,

etc. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring certain aspects.

(10) Reference throughout this specification to "one embodiment" or "an embodiment" means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the phrases "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

(11) Embodiments disclosed herein describe techniques for detecting and validating the presence of a package attached to a UAV providing aerial delivery services. Detection and validation of an attached package not only includes determining whether any package is attached, not attached, or fallen off, but may also include ensuring/validating that the package is the correct package, is correctly attached or mounted during flight, and ensuring that the correct package is attached/detached at the appropriate times throughout the various mission segments (e.g., hover segment for pickup, cruise segment enroute to the drop-off destination, hover segment during drop-off, cruise segment on return, and hover segment for landing back at the nest).

(12) The techniques described use a visual perception signal (e.g., machine vision) from capturing an image of a scene below the airframe of the UAV to identify the presence (or lack thereof) of a package. In some embodiments, image segmentation is used to parse the image and tag image pixels deemed to satisfy a package classification (e.g., texture, color, etc. of the pixels are recognizable as a package or box). However, at times the visual perception signal may not be sufficient. In particular, false negatives (i.e., package not recognized) may temporarily arise due to adverse exposure conditions, adverse backgrounds, etc. Accordingly, in various embodiments, the visual perception signal is combined with tactile perception signals from on-board sensors, including winch sensors that monitor operation of a winch that raises and lowers the package on a tether. The tether is dispensed below the UAV during pickups and drop-offs (e.g., hover segments of an aerial delivery mission). For example, these tactile perception signals may include outputs from winch sensors that monitor a length of the tether dispensed, monitor an acceleration of dispensing the tether, monitor a velocity of dispensing the tether, monitor a line tension on the tether, as well as, outputs from other sensors (e.g., GPS, altimeter, anemometer) may be referenced and combined with feedback from the visual perception and tactile perception signals to reduce false negatives and improve overall accuracy of package detection and validation.

(13) These hybrid visual and tactile perception techniques for package detection and validation can help improve the reliability, efficiency, and safety of an aerial delivery service. For example, accurate, real-time package detection/validation can ensure the correct package is picked up at the nest to reduce instances of erroneous packages being delivered. In high wind scenarios, packages can dangle, twist, or lodge into incorrect positions. Package detection/validation can be used to ensure that a picked-up package is correctly attached and cradled to the underbelly of the UAV before embarking on a cruise segment of an aerial delivery mission. While raising or lowering a package at a delivery destination, packages or the tether may catch on ground-based obstructions (e.g., tree branches, wires, etc.). Detecting these errors can help the UAV make remedial flight mission decisions. Upon delivery of a package, a customer may unexpectedly attach a return package to the tether. In these scenarios, an unexpected package should be rejected by releasing the tether so as not to return the unexpected package to the nest. The hybrid visual and tactile perception techniques described herein for package detection/validation can identify these and other scenarios, and provide valuable feedback for making real-time flight mission decisions.

(14) FIGS. **1**A and **1**B illustrate a UAV **100** adapted for executing aerial delivery missions, in accordance with an embodiment of the disclosure. FIG. **1**A is a topside perspective view illustration of UAV **100** while FIG. **1**B is a bottom side plan view of the same.

(15) The illustrated embodiment of UAV **100** is a vertical takeoff and landing (VTOL) UAV that

includes separate propulsion units **106** and **112** for providing horizontal and vertical propulsion, respectively. UAV **100** is a fixed-wing aerial vehicle, which as the name implies, has a wing assembly **102** that can generate lift based on the wing shape and the vehicle's forward airspeed when propelled horizontally by propulsion units **106**. The illustrated embodiment of UAV **100** has an airframe that includes a fuselage **104** and wing assembly **102**. In one embodiment, fuselage **104** is modular and includes a battery module, an avionics module, and a mission payload module. These modules are secured together to form the fuselage or main body.

(16) The battery module (e.g., fore portion of fuselage **104**) includes a cavity for housing one or more batteries for powering UAV **100**. The avionics module (e.g., aft portion of fuselage **104**) houses a control system including flight control circuitry of UAV **100**, which may include a processor and memory, communication electronics and antennas (e.g., cellular transceiver, wife transceiver, etc.), and various sensors (e.g., global positioning sensor, an inertial measurement unit, a magnetic compass, a radio frequency identifier reader, etc.). The mission payload module (e.g., middle portion of fuselage **104**) houses equipment associated with a mission of UAV **100**. For example, the mission payload module may include a winch **115** (see FIG. **2**B) for holding and releasing an externally attached payload (e.g., package for delivery) by dispensing a tether comprising a line and end receptacle **116** for attaching to packages. In the illustrated embodiment, end receptacle **116** has a shape that mates into a recess **117** in the underside of UAV **100** to stabilize and cradle the packages during cruise segments.

(17) In some embodiments, the mission payload module may include camera/sensor equipment (e.g., camera, lenses, radar, lidar, pollution monitoring sensors, weather monitoring sensors, scanners, etc.). In FIG. **2**B, an onboard camera **120** is mounted to the underside of UAV **100** to support a computer vision system for visual triangulation/navigation and capture images of the scene below UAV **100** for package detection and validation. In some embodiments, camera **120** may also operate as an optical code scanner for reading visual codes affixed to packages. These visual codes may be associated with or otherwise match to delivery missions and provide the UAV with a handle for accessing destination, delivery, and package validation information.

(18) As illustrated, UAV **100** includes horizontal propulsion units **106** positioned on wing assembly **102** for propelling UAV **100** horizontally. UAV **100** further includes two boom assemblies **110** that secure to wing assembly **102**. Vertical propulsion units **112** are mounted to boom assemblies **110**. Vertical propulsion units **112** providing vertical propulsion. Vertical propulsion units **112** may be used during a hover mode/segment where UAV **100** is descending (e.g., to a delivery location), ascending (e.g., at initial launch or following a delivery), or maintaining a constant altitude. Stabilizers **108** (or tails) may be included with UAV **100** to control pitch and stabilize the aerial vehicle's yaw (left or right turns) during cruise. In some embodiments, during a cruise mode/segment vertical propulsion units **112** are disabled or powered low and during a hover mode/segment horizontal propulsion units **106** are disabled or powered low.

(19) Many variations on the illustrated fixed-wing aerial vehicle are possible. For instance, aerial vehicles with more wings (e.g., an “x-wing” configuration with four wings), are also possible. Although FIGS. **1**A and **1**B illustrate one wing assembly **102**, two boom assemblies **110**, two horizontal propulsion units **106**, and six vertical propulsion units **112** per boom assembly **110**, it should be appreciated that other variants of UAV **100** may be implemented with more or less of these components.

(20) It should be understood that references herein to an “unmanned” aerial vehicle or UAV can apply equally to autonomous and semi-autonomous aerial vehicles. In a fully autonomous implementation, all functionality of the aerial vehicle is automated; e.g., pre-programmed or controlled via real-time computer functionality that responds to input from various sensors and/or pre-determined information. In a semi-autonomous implementation, some functions of an aerial vehicle may be controlled by a human operator, while other functions are carried out autonomously. Further, in some embodiments, a UAV may be configured to allow a remote

operator to take over functions that can otherwise be controlled autonomously by the UAV. Yet further, a given type of function may be controlled remotely at one level of abstraction and performed autonomously at another level of abstraction. For example, a remote operator may control high level navigation decisions for a UAV, such as specifying that the UAV should travel from one location to another (e.g., from a warehouse in a suburban area to a delivery address in a nearby city), while the UAV's navigation system autonomously controls more fine-grained navigation decisions, such as the specific route to take between the two locations, specific flight controls to achieve the route and avoid obstacles while navigating the route, and so on.
(21) FIG. **2** is a functional block diagram illustrating subsystems of a UAV (e.g., UAV **100**) relevant for detecting and validating the presence of a package, in accordance with an embodiment of the disclosure. These subsystems include a control system **205**, a camera **210**, winch sensors **215**, an anemometer **220**, other sensors **225**, and a winch **230**. The illustrated embodiment of control system **205** includes an image analysis module **235**, package presence module **240**, mission decision module **245**, and navigation module **250**. Winch **115** and camera **120** illustrated in FIG. **1**A are possible implementations of winch **230** and camera(s) **210**, respectively.
(22) Winch **230** (or **115**) is used to raise and lower packages via dispensing a tether. The packages can be clipped onto the tether at a first end and the other end coils around winch **230**, which in turn is mounted to the airframe of UAV **100**. Winch **230** enables UAV **100** to retrieve and deliver packages while hovering a few meters above the ground keeping people out of harm's way. Winch sensors **215** may be positioned to monitor operation of winch **230**. For example, a rotary encoder, or otherwise, may be used to monitor the length of the tether dispensed. The rotary encoder may also be used to measure and monitor velocity or acceleration of dispensing the tether. Velocity and/or acceleration may be monitored to identify snags or catches of the package during transport. Force feedback sensors or motor current sensors may be used to measure line tension on the tether. Line tension may be monitored as a validation that a package has been attached to the tether, but also to confirm that the correct package has been attached based upon the package weight, and thus the line tension falling within an expected range associated with the expected/correct package weight. The winch sensors **215** may collectively be referred to as tactile sensors that provide a tactile perception of whether a package (or the correct package) is attached to UAV **100**.
(23) Camera(s) **210** are mounted to the airframe of UAV **100** and oriented to capture an image of a scene below the airframe. In one embodiment, camera(s) **210** may be implemented as a single camera that is used for both visual detection of packages and visual navigation of UAV **100**. In other embodiments, multiple cameras **210** may be mounted to UAV **100** with each independently dedicated to package detection, visual navigation, or other visual tasks.
(24) Image analysis module **235** receives images acquired by camera **210** and analyzes those pictures to identify whether a package is present. In some embodiments, image analysis module **235** may also determine whether the correct package is attached by analyzing size, geometric arrangement (e.g., shape, etc.), color, position, affixed bar codes, or other visual characteristics. In yet other embodiments, image analysis module **235** may also determine whether the package has been correctly attached/stowed for a cruise segment based upon the location of the package in the image.
(25) Image analysis module **235** may be implemented using a variety of different technologies. In one embodiment, image analysis module **235** is a digital image processor that performs “image segmentation” to identify different objects and their respective boundaries in the image. Image segmentation is also referred to as “semantic segmentation” as the individual objects are semantically classified into object categories (e.g., package, tree, building, pole, etc.). Accordingly, the process of image segmentation tags pixels in the image as belonging to a particular classification. In the case of package detection, image analysis module **235** tags pixels that are deemed to satisfy a “package classification.” FIG. **4**A illustrates an example image **400** of a scene below UAV **100** while UAV **100** is in a cruise segment of its aerial delivery mission enroute to a

delivery destination. As can be seen, a package **405** is present in image **400**. In fact, just a corner of package **405** extends from a side of the image while a majority of package **405** is out-of-frame of image **400**. This is a typical arrangement for package **405** when it is property attached and cradled to UAV **100** enroute to its delivery destination during a cruise segment. FIG. **4**B illustrates an image or semantic segmentation that identifies a contiguous group **410** of pixels all tagged as satisfying the package classification.

(26) When a threshold number of contiguous pixels are tagged as being a member of the package classification, then image analysis module **235** determines that a package is viewable in the image. This threshold number may have a variable threshold dependent upon the length of the tether dispensed when the image was acquired. In other words, if the tether is fully recoiled (e.g., see FIGS. **4**A and **4**B), the threshold number of pixels (e.g., size of contiguous pixels) is different than when the tether **415** is fully extended (e.g., see image **401** in FIG. **4**C). Similarly, the identifiable shape and expected position of the contiguous pixels (i.e., the package **420**) should be different based on the length of the tether **415** dispensed when the image is acquired. The presence of wind or crosswind is also expected to affect the orientation and position of the package **420** in image **401**.

(27) As mentioned, image analysis module **235** may be implemented using a variety of different technologies. In other embodiments, image analysis module **235** may implement a bounding box classification (also referred to as an object detection classification) or otherwise. In one embodiment, image analysis module **235** may be implemented with a neural network or machine learning (ML) classifier trained on a dataset of images that have been previously labeled with ground truth data. In other words, the ML classifier may be trained to perform image segmentation, bounding box classification, or otherwise, as discussed above.

(28) Package presence module **240** receives visual perception data **212** from image analysis module **235** along with tactile perception data **217** from winch sensors **215**, and in some embodiments, may also receive sensor data from anemometer **220** and other sensors **225**. The other sensors **225** may include a GPS sensor, an altimeter sensor, or otherwise. The other sensor data, anemometer data, tactile perception data **217**, and visual perception data **212** may all be analyzed to provide package presence module **240** a wholistic, multifaceted perception of the immediate environment directly impacting the package to make a more informed decision as to whether a package is presently attached to the tether, whether the correct package is attached, or whether the package is attached in the correct position. Package presence module **240** may be implemented as a heuristic algorithm, a trained ML model, or a combination of both. In the case of a ML model or neural network, sensor data correlated to images from past aerial delivery missions representative of different real world scenarios may be used to train package presence module **240**.

(29) Some example scenarios are now described. If UAV **100** currently believes it is holding package **405** and is operating along a cruise segment enroute to delivery, but suddenly the tether experiences rapid acceleration, high velocity, and increasing tether length coupled with a rapidly changing number of pixels satisfying the package classification, then package presence module **240** may deem the package has been snagged and no longer correctly attached. If the shape of the package or the position of the package is not as expected, these deviations may be interpreted as a twisted or faulty attachment of the package. Characteristic patterns of tether length, acceleration, velocity, altimeter, and GPS position relative to pickup/delivery locations present themselves during hover and cruise segments of an aerial delivery mission. Deviations from these characteristic patterns in the visual and/or tactile perception data can be interpreted to represent package errors such as no package, incorrect package, package mounted wrong, etc. In the illustrated embodiment. package presence module **240** is responsible for making these package detection/validation decision based upon the visual perception data **212**, tactile perception data **217**, and other sensor data.

(30) Mission decision module **245** includes yet another level of logic which uses output from the

package presence module **240** to make real-time flight mission decisions. For example, mission decision module **245** may decide to continue on with a mission, if the package presence module **240** determines that the correct package is correctly attached, while it may decide to cancel the mission if any of the above discussed package errors arise. Accordingly, mission decision module **245** understands the various mission segments and when a package is expected to be attached or dropped off in relation to the mission segments. For example, if a package is delivered at a drop-off location while UAV **100** is in a hover segment, but suddenly package presence module **240** detects the presence of a new unexpected package (i.e., the customer removed the delivery package from the tether but then attached a new unexpected package for return, then mission decision module **245** can react accordingly. One such response may be to release the tether and return to the nest without the unexpected package. In this example, unexpected packages are deemed unsafe and not returned to the nest. Similarly, if the package is deemed to be lost unrouted to a delivery destination, mission decision module **245** can make the inflight decision to cancel the aerial delivery mission and return to the nest without flying to the drop-off destination empty handed.
(31) As illustrated, navigation module **250** can take input from cameras **210** and mission decision module **245** to plan and revise routes in real-time. Navigation module **250** may receive input from other sensors **225** using GPS signals for navigation. However, it is also noteworthy that navigation module **250** may also include visual navigation logic to navigate based off the images output from camera(s) **210**. In one embodiment, image analysis module **235** and the visual navigation logic share the same camera **210**.
(32) FIG. **3** is a flow chart illustrating a process **300** for detecting and/or validating the presence of a package attached to UAV **100** using a combination of visual image analysis and tactile feedback from sensors, in accordance with an embodiment of the disclosure. The order in which some or all of the process blocks appear in process **300** should not be deemed limiting. Rather, one of ordinary skill in the art having the benefit of the present disclosure will understand that some of the process blocks may be executed in a variety of orders not illustrated, or even in parallel.
(33) Upon commencement of an aerial delivery mission (process block **305**), control system **205** begins gathering visual perception data **212** based on images of the scene below UAV **100** (process block **310**), tactile perception data **217** from winch sensors **215** (process block **315**), and other sensor data from other sensors **225** (process block **320**) and analyzing this multifaceted data in the context of its current mission segment. A determination of its current mission segment (decision block **325**) may be made based upon GPS data and/or navigation data from navigation module **250**.
(34) Image analysis module **235** analyzes the acquired images using image segmentation (process block **330**) to determine whether a package is attached (process block **335**). Module **235** provides its visual perception determinations based solely on the images acquired by cameras **210** and passes these determinations onto package presence module **240**. In addition to package present or not present, the visual perception determinations may also include visual perception determinations of whether the correct package is attached or whether the package is positioned correctly.
(35) Module **240** collects both the visual perception data **212** along with the tactile perception data **217** to make the ultimate determination of whether a package is attached (process block **340**), whether the package is the correct package and/or an expected package (process block **345**), or whether the package is correctly attached in the expected location (process block **350**). Accordingly, module **240** relies upon both types of data along with sensor data from other sensors **225** and/or anemometer **220** to make a more accurate determination with fewer false negatives as to whether a package is present, whether it is the correct/expected package, or whether it is correctly attached to UAV **100**.
(36) Visual perception data **212** may also be used as a built-in self-test to validate the other subsystems (process block **355**). The package size and position as determined by module **235** using image segmentation on images captured by camera **210** can be used to validate correct operation of winch **230**, winch sensors **215**, or anemometer **220**. For example, the line length measured by a

rotary encoder can be validated against the package size measured in the image, which corresponds to a visual estimate of how far the package is from UAV **100** and camera **210**. If anemometer **220** measures a certain crosswind, then image analysis along with the current tether length can be used to cross-check whether the package is swinging in the expected direction by an expected amount.
(37) Finally, mission decision module **245** can use the package presence/validation determination output from module **240** along with its knowledge of the current mission segment to make real-time flight mission decisions that can change based upon whether the package is present or not present as expected (process block **360**). These flight mission decisions can change the next destination or waypoint, or result in a mission abort (process block **365**).
(38) The processes explained above are described in terms of computer software and hardware. The techniques described may constitute machine-executable instructions embodied within a tangible or non-transitory machine (e.g., computer) readable storage medium, that when executed by a machine will cause the machine to perform the operations described. Additionally, the processes may be embodied within hardware, such as an application specific integrated circuit ("ASIC") or otherwise.
(39) A tangible machine-readable storage medium includes any mechanism that provides (i.e., stores) information in a non-transitory form accessible by a machine (e.g., a computer, network device, personal digital assistant, manufacturing tool, any device with a set of one or more processors, etc.). For example, a machine-readable storage medium includes recordable/non-recordable media (e.g., read only memory (ROM), random access memory (RAM), magnetic disk storage media, optical storage media, flash memory devices, etc.).
(40) The above description of illustrated embodiments of the invention, including what is described in the Abstract, is not intended to be exhaustive or to limit the invention to the precise forms disclosed. While specific embodiments of, and examples for, the invention are described herein for illustrative purposes, various modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize.
(41) These modifications can be made to the invention in light of the above detailed description. The terms used in the following claims should not be construed to limit the invention to the specific embodiments disclosed in the specification. Rather, the scope of the invention is to be determined entirely by the following claims, which are to be construed in accordance with established doctrines of claim interpretation.

Claims

1. An unmanned aerial vehicle (UAV) adapted for package deliveries, the UAV comprising: an airframe adapted for carrying a package; a camera mounted to the airframe and oriented to capture an image of a scene below the airframe; a tether mounted to the airframe at a first end and adapted to attach the package to the UAV via a second end opposite the first end; and a control system coupled to the camera and including logic, that when executed by the control system, causes the UAV to perform operations including: capturing the image of the scene below the airframe; analyzing the image to classify pixels in the image deemed to satisfy a package classification; and determining whether the package is attached to the UAV based at least on whether a threshold number of contiguous pixels in the image are classified into the package classification.
2. The UAV of claim 1, wherein analyzing the image comprises: performing an image segmentation on the image; and tagging the pixels in the image deemed to satisfy the package classification.
3. The UAV of claim 1, wherein determining whether the package is attached to the UAV further comprises: determining whether the threshold number of contiguous pixels classified into the package classification are positioned at an expected location within the image.
4. The UAV of claim 2, wherein determining whether the package is attached to the UAV is further based on at least one of a geometric arrangement of the contiguous pixels in the image or a position

of the contiguous pixels in the image.

5. An unmanned aerial vehicle (UAV) adapted for package deliveries, the UAV comprising: an airframe adapted for carrying a package; a camera mounted to the airframe and oriented to capture an image of a scene below the airframe; a tether mounted to the airframe at a first end and adapted to attach the package to the UAV via a second end opposite the first end; a winch mounted to the airframe and adapted for raising or lowering the package to or from the UAV via the tether; and a control system coupled to the camera and the winch, the control system including logic, that when executed by the control system, causes the UAV to perform operations including: capturing the image of the scene below the airframe; tracking a length of the tether dispensed by the winch; analyzing the image; and determining whether the package is attached to the UAV based at least on whether the analyzing indicates that a representation of the package in the image exceeds a threshold, wherein the threshold is variable based on the length of the tether dispensed when the image is acquired.

6. The UAV of claim 5, wherein determining whether the package is attached to the UAV is further based on at least one of an acceleration or a velocity of dispensing the tether.

7. The UAV of claim 5, wherein determining whether the package is attached to the UAV is further based on a line tension on the tether.

8. The UAV of claim 7, wherein determining whether the package is attached to the UAV includes determining whether a correct package is attached to the UAV, wherein the control system includes additional logic, that when executed by the control system, causes the UAV to perform additional operations including: determining whether the line tension falls within an expected range associated with the correct package.

9. The UAV of claim 5, wherein the control system includes additional logic, that when executed by the control system, causes the UAV to perform additional operations including: making a first flight mission decision based upon determining the package is attached to the UAV and the package is expected to be attached to the UAV; and making a second flight mission decision, different than the first flight mission decision, based upon determining the package is attached to the UAV and the package is not expected to be attached to the UAV.

10. The UAV of claim 9, wherein the second flight mission decision comprises releasing the tether from the UAV.

11. The UAV of claim 5, wherein analyzing the image to identify whether the package is present in the image comprises: performing an image segmentation on the image; tagging pixels in the image deemed to satisfy a package classification; and determining the package is viewable in the image of the scene below the airframe based at least on whether a threshold number of contiguous pixels in the image are tagged as satisfying the package classification, wherein the threshold number of contiguous pixels is a variable threshold dependent upon the length of the tether dispensed when the image is acquired.

12. The UAV of claim 5, wherein the control system includes additional logic, that when executed by the control system, causes the UAV to perform additional operations including: analyzing the image to estimate how far the package is from the UAV; and validating correct operation of the winch or winch sensors that monitor operation of the winch based at least on the image and the estimate.

13. The UAV of claim 1, wherein the control system includes additional logic, that when executed by the control system, causes the UAV to perform additional operations including: generating navigation decisions for the UAV based at least upon the image of the scene below the airframe acquired by the camera, wherein visual navigation and package presence decisions are both based upon images acquired by the camera.

14. A method of validating a presence of a package carried by an unmanned aerial vehicle (UAV), the method comprising: capturing an image of a scene below the UAV with a camera mounted to the UAV and oriented to face down from the UAV; analyzing the image to classify pixels in the

image deemed to satisfy a package classification; and determining whether the package is attached to the UAV via a tether extending from an underside of the UAV, based at least on whether a threshold number of contiguous pixels in the image are classified into the package classification.
15. The method of claim 14, wherein determining whether the package is attached to the UAV includes determining whether a corner of the package extends from a side of the image while a majority of the package is out-of-frame of the image when the UAV is operating in a cruise segment of an aerial delivery mission.
16. The method of claim 14, wherein analyzing the image comprises: performing an image segmentation on the image; and tagging the pixels in the image deemed to satisfy the package classification.
17. The method of claim 14, wherein determining whether the package is attached to the UAV further comprises: determining whether the threshold number of contiguous pixels classified into the package classification are positioned at an expected location within the image, the expected location changing dependent upon a mission segment of an aerial delivery mission flown by the UAV.
18. The method of claim 14, further comprising: raising or lowering the package via a winch coupled to the tether; and tracking a length of the tether dispensed by the winch, wherein determining whether the package is attached to the UAV is based at least on the analyzing of the image and the length of the tether dispensed when the image is acquired.
19. The method of claim 18, wherein determining whether the package is attached to the UAV is further based on at least one of an acceleration of dispensing the tether, a velocity of dispensing the tether, or a line tension on the tether.
20. The method of claim 14, wherein determining whether the package is attached to the UAV includes determining whether a correct package is attached to the UAV, the method further comprising: determining whether a line tension of the tether falls within an expected range associated with the correct package.
21. A method of validating a presence of a package carried by an unmanned aerial vehicle (UAV), the method comprising: capturing an image of a scene below the UAV with a camera mounted to the UAV and oriented to face down from the UAV; analyzing the image to identify whether the package is present in the image; determining whether the package is attached to the UAV, via a tether extending from an underside of the UAV, based at least on the analyzing of the image; and releasing the tether when determining that an incorrect package or unexpected package has been attached to the UAV after delivering the package to a destination.